Modeling electrical losses in CsPbI3 solar cells: stacked idealizations and a routine-quality envelope at 18.0%

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Abstract. Stacked idealisations inflate SCAPS-1D CsPbI3 projections. On one fully worked CsPbI3/CBTS stack, joint optimisation of thickness, doping, defects, gap, and both transport-layer affinities gives a permissive-box ceiling of **24.95%**. Sampling a frozen routine-quality prior volume (96-point Latin hypercube over eight factors; ranges and measured anchors in Table 1 of the manuscript) gives a median of **18.0%** with a 5–95% prior-predictive envelope of **15.6–20.0%**. This is a sensitivity envelope under stated priors, not a confidence interval on real devices. The 2024–2026 experimental records fall above that envelope. The ceiling record comparison is density-of-states conditional: with capped density of states, literature radiative recombination ($C_r = 10^{-10}$ cm^3/s) lowers the ceiling to 21.86%, below the 22.02% certified record; with effective-mass physical density of states, the same radiative term gives 23.46% ($N_v = 4.1 \times 10^{18}$ cm^{-3}) and 23.04% ($N_v = 8.9 \times 10^{18}$ cm^{-3}), above it. Interface sensitivity is regime dependent, bulk defects dominate at routine defect density, and series resistance up to 5 Ωcm^2 closes roughly half of the six-point fill-factor gap against measured devices, with the remainder attributed to nonradiative interface recombination (ideality above unity). (199 words; one paragraph; no displayed equations, references, or tables.)

Keywords: CsPbI3; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

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Conflict of interest. The authors have no conflicts to disclose. (Verbatim AIP statement; to be inserted in the manuscript at revision — see README.)

Data availability. The data that support the findings of this study are openly available in the public GitHub repository (definition files, sweep receipts, raw curves, scripts, verification receipts, and typeset sources). See the Data availability section of the manuscript for the URL and revision. Copyrighted reference PDFs are listed, not redistributed.